

Mathematics A

Unit 2 • Chapter OL-T42 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

3 classified items • 14 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T42 • Expertise Q16+ • 3 items • 14 marks

Chapter	Items	Marks	Starts
01. OL-T42 Circle Theorems	3	14	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Circle Theorems

Unit 2 • Expertise • 14 marks • 3 item(s)

November 2023 | 4MA1-1H Q22 | 6 marks

Use radius perpendicular to tangent

November 2024 | 4MA1-2H Q16 | 5 marks

Complete a multi-theorem circle-angle chain

January 2022 | 4MA1-2H Q19 | 3 marks

Use the internal intersecting-chords product

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q22 demand

6 marks

- 22 [In this question 1 cm = 1 unit on the x -axis and
1 cm = 1 unit on the y -axis]

P is a point on a circle with centre $(0, 0)$

The coordinates of P are $(8, -10)$

The line L is the tangent to the circle at the point P

L crosses the x -axis at the point Q and crosses the y -axis at the point R

Work out the length of RQ

Give your answer correct to 3 significant figures.

..... cm

WORKING SPACE

4MA1-1H Q22 demand

6 marks

- 22 [In this question 1 cm = 1 unit on the x -axis and
1 cm = 1 unit on the y -axis]

P is a point on a circle with centre $(0, 0)$

The coordinates of P are $(8, -10)$

The line L is the tangent to the circle at the point P

L crosses the x -axis at the point Q and crosses the y -axis at the point R

Work out the length of RQ

Give your answer correct to 3 significant figures.

..... cm

Length between tangent intercepts

6 marks

Demand – Length between tangent intercepts

Find radius gradient

$$m_{OP} = (-10 - 0)/(8 - 0) = -5/4.$$

Find tangent gradient

$$m_L = 4/5 \text{ since } m_{OP}m_L = -1.$$

Find intercepts

$$y + 10 = \frac{4}{5}(x - 8) \Rightarrow R = (0, -16.4), Q = (20.5, 0).$$

Distance

$$RQ = \sqrt{20.5^2 + 16.4^2} = 26.3 \text{ cm (3 s.f.)}.$$

Final answer

26.3 cm

4MA1-2H Q16 M01, M02, M03, M04

5 marks

16 A, B, C and D are points on a circle, centre O

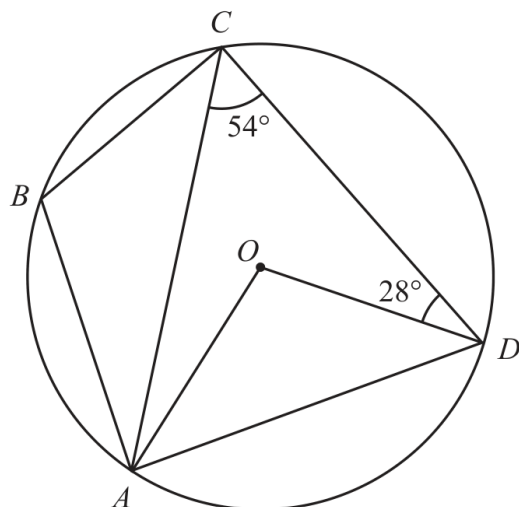


Diagram **NOT**
accurately drawn

(a) (i) Work out the size of angle AOD

.....
(1)

(ii) Give a reason for your answer to part (a)(i)

.....
.....
(1)

(b) Work out the size of angle CAO

.....
(1)

(c) Work out the size of angle ABC

.....
(2)

Continue your method**5 marks**

WORKING SPACE

4MA1-2H Q16 M01, M02, M03, M04

5 marks

16 A, B, C and D are points on a circle, centre O

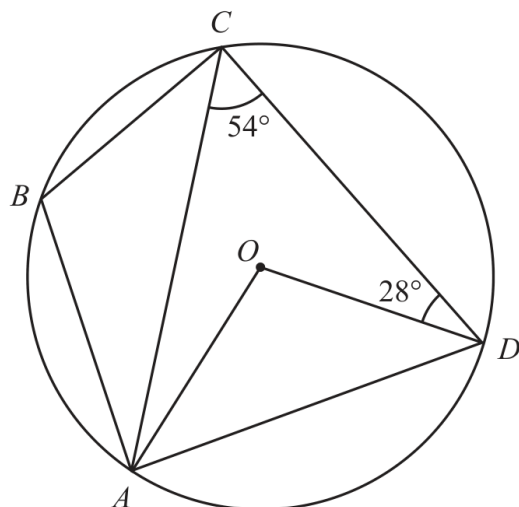


Diagram **NOT**
accurately drawn

(a) (i) Work out the size of angle AOD

.....
(1)

(ii) Give a reason for your answer to part (a)(i)

.....
.....
(1)

(b) Work out the size of angle CAO

.....
(1)

(c) Work out the size of angle ABC

.....
(2)

Find a central angle / Give the circle-theorem reason / Find angle CAO / Find angle ABC

5 marks

M01 – Find a central angle**Use the central-angle theorem**

$$\angle AOD = 2(54^\circ) = 108^\circ.$$

Final answer

108 degrees

M02 – Give the circle-theorem reason**State the theorem**

The angle at the centre is twice the angle at the circumference standing on the same arc.

Final answer

Angle at centre is twice angle at circumference

M03 – Find angle CAO**Use equal radii**OA=OC, so the base angles in triangle AOC are equal; with the central angle, $\angle CAO=26^\circ$.**Final answer**

26 degrees

M04 – Find angle ABC**Find the adjacent angle**

Use the radius/chord angles to obtain the angle at C in triangle ABC.

Use angles in a triangle

$$\angle ABC = 116^\circ.$$

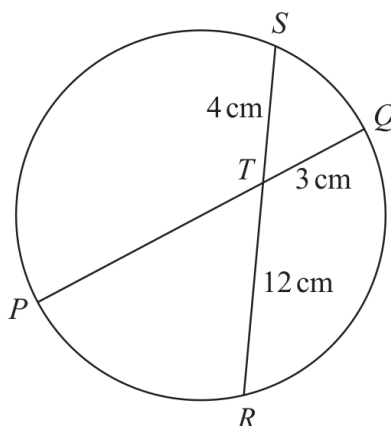
Final answer

116 degrees

4MA1-2H Q19

3 marks

19

Diagram **NOT**
accurately drawn

PTQ is a diameter of a circle.
 RTS is a chord of the circle.

$$TQ = 3 \text{ cm} \qquad ST = 4 \text{ cm} \qquad TR = 12 \text{ cm}$$

Calculate the radius of the circle.

..... cm

Continue your method

3 marks

WORKING SPACE

4MA1-2H Q19

3 marks

19

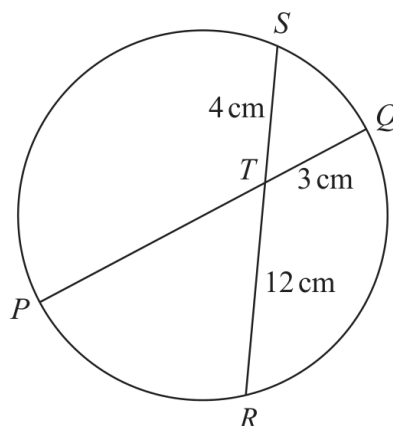


Diagram **NOT** accurately drawn

PTQ is a diameter of a circle.
 RTS is a chord of the circle.

$$TQ = 3\text{ cm} \quad ST = 4\text{ cm} \quad TR = 12\text{ cm}$$

Calculate the radius of the circle.

..... cm

Chapter: Circle Theorems

3 marks

Q19 – Chapter: Circle Theorems

Family: Use intersecting chord and secant products • Variant: Use the internal intersecting-chords product

MethodBy intersecting chords, $PT \times TQ = ST \times TR$, so $PT \times 3 = 4 \times 12$ and $PT = 16$ cm.*Why: Use the chord-intersection theorem.***Method** $PQ = PT + TQ = 16 + 3 = 19$ cm.*Why: P, T, Q are collinear and PQ is a diameter.***Method**Radius = $PQ/2 = 9.5$ cm.*Why: A radius is half a diameter.***Final answer**

radius: 9.5 cm